
TENSOR-NETWORK BASED INFERENCE FOR THE EVENT CALCULUS

RESEARCH NOTES IN THE ENEXA AND QROM PROJECTS

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ABSTRACT

We investigate the tensor-based inference scheme in Tsilionis et al. (2024) exploiting the law of inertia in the event calculus. We derive representation of the laws of inertia as tensor network contraction equations exploiting the basis calculus scheme in Goessmann et al. (2026).

1 Event calculus

We introduce two term variables:

- f fluent, enumerated by O_f
- t time stamps, enumerated by O_t

Further the function:

- $suc(t)$ successor of the time stamp

And the predicates:

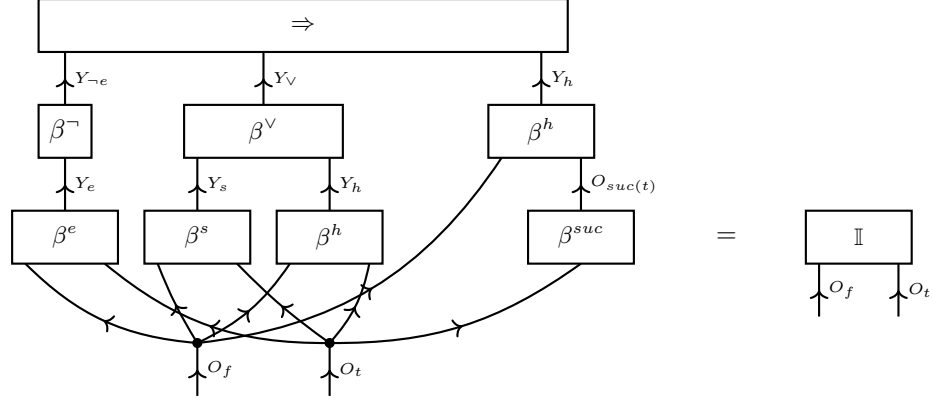
- $h(f, t)$ fluent f holds at time stamp t
- $e(f, t)$ fluent f ends at time stamp t
- $s(f, t)$ fluent f starts at time stamp t

2 Laws of inertia

"If a fluent holds or starts at a time t , and it is not terminated at time t , then it holds also at the succeeding time stamp $suc(t)$:"

$$((h(f, t) \vee s(f, t)) \wedge \neg e(f, t)) \rightarrow h(f, suc(t))$$

Exploiting basis calculus, these pose the contraction equation on the basis encodings of the predicates and functions (which specify the FOL worlds):



When choosing the time order on the time stamps, the basis encoding of suc is (called U in Tsilionis et al. (2024))

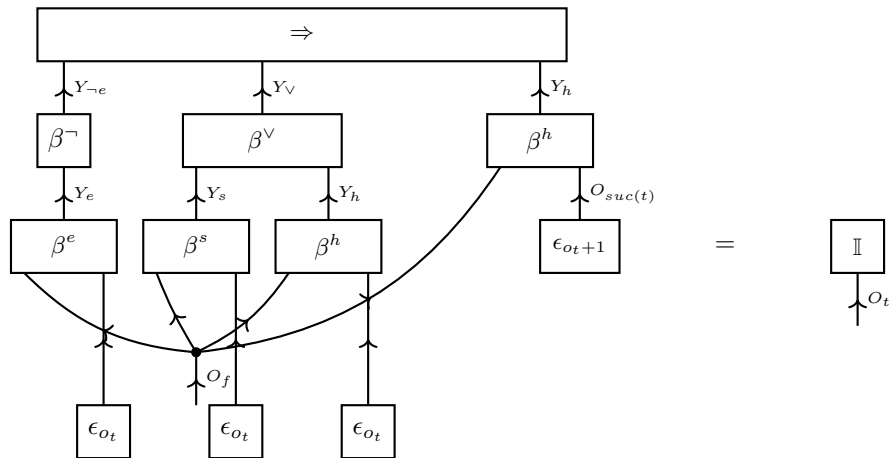
$$\beta^{suc} [O_{suc(t)} = o_{suc(t)}, O_t = o_t] = \begin{cases} 1 & \text{if } o_{suc(t)} = o_t + 1 \\ 0 & \text{else.} \end{cases}$$

3 Inference

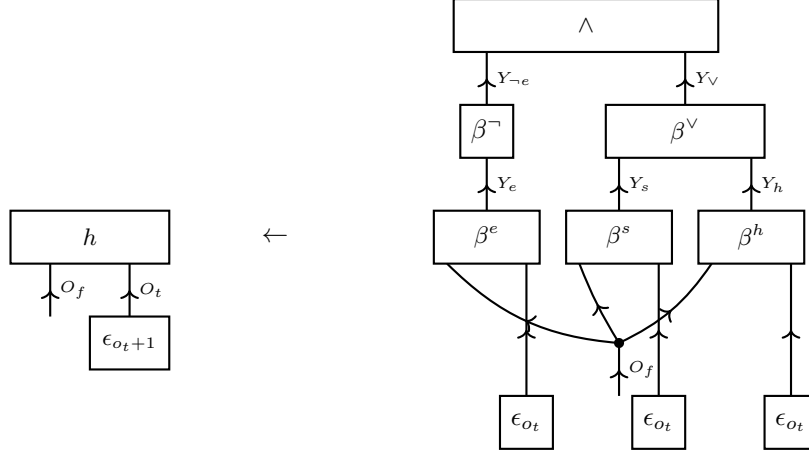
Let us consider that the groundings to the predicates $s[O_f, O_t]$ and $t[O_f, O_t]$ and to the function suc are known, and consider inferring the least tensor $h[O_f, O_t]$ respecting the laws of inertia:

$$\text{argmin}_{h[O_f, O_t]} \langle h[O_f, O_t] \rangle_{[\emptyset]} \quad \text{subject to the laws of inertia holding for given } s[O_f, O_t], t[O_f, O_t]$$

To solve the contraction equation we contract with any ϵ_{o_t} for $o_t \in [r_t]$ to get the equation:



We solve them iteratively by computing the vectors on the right hand side and write them into the slices of h as:



Note that we exploit in the solution the consecutive dependence structure of the equations, induced by the β^{suc} matrix.

4 Generic solution strategies

The solution of contraction equations can in general be done by simplifying them:

- Close some open legs with arbitrary tensors: When choosing basis tensors (or more general normed tensors), the right hand side is still a $\mathbb{I}$ tensor. We used this trick to derive sequences of simpler equations for each time stamp.
- Contract only a subset of the left hand side: We could restrict by removing β^s and β^h to the two original laws of inertia in Tsilionis et al. (2024).

References

- Alex Goessmann, Janina Schütte, Maximilian Fröhlich, and Martin Eigel. A tensor network formalism for neuro-symbolic AI, January 2026. URL <http://arxiv.org/abs/2601.15442>. arXiv:2601.15442 [cs].
- Efthimis Tsilionis, Alexander Artikis, and Georgios Paliouras. A Tensor-Based Formalization of the Event Calculus. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*, pages 3584–3592, Jeju, South Korea, August 2024. International Joint Conferences on Artificial Intelligence Organization. ISBN 978-1-956792-04-1. doi: 10.24963/ijcai.2024/397. URL <https://www.ijcai.org/proceedings/2024/397>.